

《计算机网络》作业

第十二次作业：（英文版教材第六章 19，22，25，26，27，28，补充题 1,补充题 2）

19.(6.5.4 The TCP Segment Header)What is the total size of the minimum TCP MTU, including TCP and IP overhead but not including data link layer overhead?

22.(6.5.4 The TCP Segment Header)In Fig. 6-36 we saw that in addition to the 32-bit acknowledgement field, there is an ACK bit in the fourth word. Does this really add anything? Why or why not?

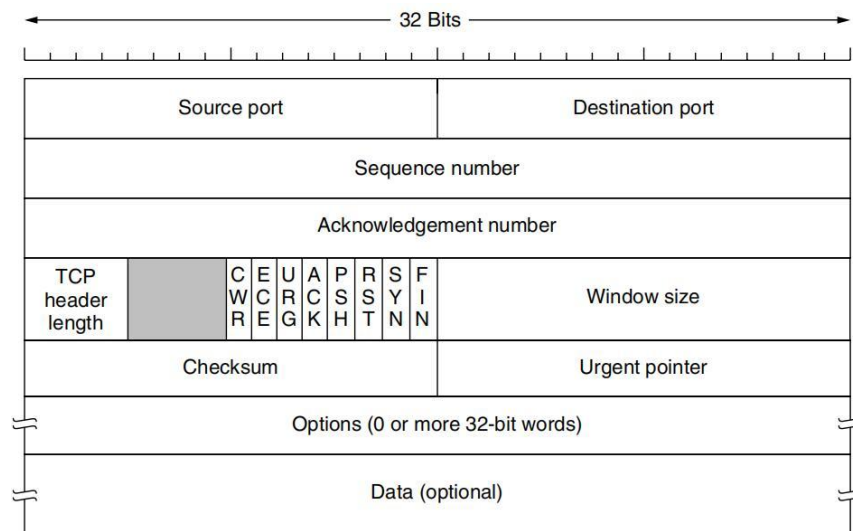


Figure 6-36. The TCP header.

25. (6.5.10 TCP Congestion Control)Consider the effect of using slow start on a line with a 10-msec round-trip time and no congestion. The receive window is 24 KB and the maximum segment size is 2 KB. How long does it take before the first full window can be sent?
26. (6.5.10 TCP Congestion Control)Suppose that the TCP congestion window is set to 18 KB and a timeout occurs. How big will the window be if the next four transmission bursts are all successful? Assume that the maximum segment size is 1 KB.
27. (6.5.9 TCP Timer Management)If the TCP round-trip time, RTT, is currently 30 msec and the following acknowledgements come in after 26, 32, and 24 msec, respectively, what is the new RTT estimate using the Jacobson algorithm? Use $\alpha = 0.9$.

28. (6.5.8 TCP Sliding Window) A TCP machine is sending full windows of 65,535 bytes over a 1-Gbps channel that has a 10-msec one-way delay. What is the maximum throughput achievable? What is the line efficiency?

补充题 1(6.5.10 TCP Congestion Control)

主机甲和乙已建立了 TCP 连接，甲始终以 MSS=1KB 大小的段发送数据，并一直有数据发送；乙每收到一个数据段都会发出一个接收窗口为 10KB 的确认段。若甲在 t 时刻发生超时并拥塞窗口为 8KB，则从 t 时刻起，不再发生超时的情况下，请计算经过 10 个 RTT 后，甲的发送窗口是多少？

补充题 2(6.5.4 The TCP Segment Header) 主机 A 向主机 B 连续发送了两个 TCP 报文段，其序号分别是 70 和 100，请问：

- (1) 第一个报文段携带了多少字节的数据？
- (2) 主机 B 收到第一个报文段后发回的确认中的确认序号应当是多少？
- (3) 主机 B 收到第二个报文段后发回的确认中的确认序号是 180，试问 A 发送的第二个报文段中的数据有多少字节？
- (4) 如果 A 发送的第一个报文段丢失了，但第二个报文段到达了 B，B 在第二个报文段到达后向 A 发送确认，这个确认的序号应为多少？

英文版教材第七章 6, 18, 20 (自己练习, 不用交本章作业)

6. (7.1 DNS) Can a machine with a single DNS name have multiple IP addresses? How could this occur?
18. (7.2.5 Final delivery) IMAP allows users to fetch and download email from a remote mailbox. Does this mean that the internal format of mailboxes has to be standardized so any IMAP program on the client side can read the mailbox on any mail server? Discuss your answer.
20. (7.2.5 Final delivery) Does Webmail use POP3, IMAP, or neither? If one of these, why was that one chosen? If neither, which one is it closer to in spirit?